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Leaf angle adjustment improves apparent covariation between fluorescence and photochemical quantum yields across leaves

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ABSTRACT

Solar-induced chlorophyll fluorescence (SIF) is commonly used to monitor photosynthetic responses to drought, but the SIF–photosynthesis relationship remains unclear due to the complex interactions between the quantum yield of photochemistry (Φ_P) and fluorescence (Φ_F), which reflect physiological energy partitioning and are indirectly influenced by structural factors through changes in absorbed light. To address this gap, we investigated the leaf angle distribution (LAD) shifts operating light regimes across leaves in sugar maple and white oak saplings, and how these shifts influence the apparent covariation between Φ_P and Φ_F across leaves using a growth-chamber drought experiment. By integrating classical techniques with remote sensing methods, we found that accounting for variation in LAD and absorbance made the apparent Φ_P-Φ_F covariation across leaves more linear. Changes in LAD altered the operating light regime across leaves, resulting in a more linear positive apparent relationship between Φ_P and Φ_F (from not significant to $R^2 = 0.6$). Additionally, we observed that the conductance of the Cytochrome b_6f complex to linear electron flow covaried with Φ_F under drought and high light conditions ($R^2 = 0.82$). Our findings suggest that considering both physiological and structural changes provides a more comprehensive understanding of the SIF-photosynthesis relationship under drought stress, particularly in relation to light energy partitioning. Focusing on only one aspect, either physiological or structural, can lead to an incomplete understanding of the mechanisms driving quantum yield relationships at the canopy scale and limit our ability to leverage SIF products to monitor how ecosystems are responding to changing environmental conditions.

1. Introduction

Temperate forests will face increased drought stress in the upcoming decades (Allen et al., 2010; Change, 2013; Costanza et al., 2023). Due to physiological regulation of stomatal conductance and structural responses like changes in leaf angle distribution (LAD), drought events can dramatically reduce photosynthesis at the leaf level and ecosystem-scale productivity (Bonan, 2008; Brando et al., 2008; Kolus et al., 2019; Novick et al., 2016). Previous studies have focused on leaf-level physiological changes to drought (e.g., stomatal conductance), but often neglect the extent to which structural changes mediate physiological responses (Saatchi et al., 2013; Smith et al., 2019; Yang et al., 2023). To understand how temperate forests will be impacted by shifting drought regimes, it is crucial to investigate the interaction between vegetation structural and physiological variables under hydrologic stress.

At the canopy scale, remote sensing has been used to monitor vegetation responses to drought stress, and new emerging technologies have facilitated advances in our understanding of plant-water relationships. Among them, solar-induced chlorophyll fluorescence (SIF) has emerged as a widely used tool to monitor photosynthesis (Frankenberg

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and Berry, 2018; Martini et al., 2022; Mohammed et al., 2019; Yang et al., 2015; Yang et al., 2026). By using satellite-based SIF as a proxy (Frankenberg et al., 2011; Joiner et al., 2013; Li and Xiao, 2019; Ryu et al., 2019; Sun et al., 2017), vegetation photosynthetic responses to drought can be detected across a myriad ecosystem types (Ač et al., 2015; Cao et al., 2021; Chen et al., 2021; Jonard et al., 2020). However, SIF and photosynthesis can be influenced by various factors, leading to diverging conclusions about their relationship. For example, the magnitude of SIF did not align with equivalent decreases in photosynthesis in a pine forest (Wohlfahrt et al., 2018) or in turfgrass (Jones et al., 2023), and one study reported a negative relationship between SIF and photosynthesis (Martini et al., 2022). Therefore, further research is needed to investigate the SIF-photosynthesis relationship during drought stress, and several gaps in our understanding still need to be addressed, which are detailed in the following paragraphs.

Plants exhibit various types of structural changes under drought, including changes in leaf area index and LAD. While leaf area index has been widely studied, LAD remains less well understood, particularly at the canopy scale. In particular, LAD changes under drought can alter leaf exposure and absorbed light across leaves, thereby influencing the apparent covariation between the quantum yields of steady-state fluorescence (Φ_F) and photochemistry (Φ_P), which are key variables related to the partitioning of light energy between photosynthesis and SIF. (Porcar-Castell et al., 2021; Porcar-Castell et al., 2014). During water stress, changes in LAD alter the orientation of leaves relative to incoming radiation. Such adjustments reduce the radiation load on the leaf surface (Anten, 2005; Nilsen and Forseth, 2018; Yang et al., 2023). Therefore, LAD adjustment indirectly modulates the apparent Φ_P-Φ_F relationship by altering the operating light regime, given that Φ_P-Φ_F covariation is fundamentally controlled by absorbed light intensity. Variation in leaf-level absorbed light directly determines the magnitudes of Φ_P and Φ_F, as these two quantities exhibit distinct light-response behaviors.

Φ_P exhibits a general decreasing trend with increasing light intensity. Light energy is mainly used for photochemical quenching (PQ) of excitation energy via electron transport and non-photochemical quenching (NPQ) during the growing season (Baker, 2008; Kaiser et al., 2019; Porcar-Castell et al., 2014). In photosystem II (PS_{II}), NPQ photoprotective mechanisms involving carotenoids of the xanthophyll cycle are upregulated with increasing light intensity to protect plant molecules (Genty et al., 1989; Kramer et al., 2004). Therefore, in general, an increase in light intensity leads to a decrease in Φ_P and an increase in the quantum yield of NPQ (Φ_{NPQ}) (Wu et al., 2022).

Φ_F demonstrates a markedly different light response compared to Φ_P. In low light conditions, Φ_F generally increases with light intensity and eventually reaches a peak before it declines with increasing light (Gu et al., 2019; van der Tol et al., 2014). The specific peak point is determined by the maximum carboxylation activity of Rubisco (V_{cmax}) and the maximum electron transport activity of Cytochrome b_6f (Johnson and Berry, 2021; Zhang et al., 2016). Consequently, Φ_F generally shows an inverted 'V' pattern in the light response curve under non-stress conditions, which significantly differs from Φ_P. Given the differences in the light responses of Φ_P and Φ_F, changes in LAD under drought stress should be considered to more accurately understand the Φ_P-Φ_F relationship.

The varying light responses of Φ_P and Φ_F result in a dynamic Φ_P-Φ_F relationship that is dependent on operating light regimes (Magney et al., 2020; Maguire et al., 2020; Porcar-Castell et al., 2021). In low light conditions (PQ-phase, Fig. 1), we expect the Φ_P-Φ_F relationship to be negative and controlled by PQ. Specifically, during this phase, PQ induces plastoquinol accumulation before engagement of NPQ dissipation, resulting in the NPQ remaining relatively constant at low levels of light. As light intensity increases (NPQ-phase, Fig. 1), we expect a positive linear Φ_P-Φ_F relationship. This is because Φ_P and Φ_F are dominated by NPQ as the change in NPQ with varying light intensity is greater than the

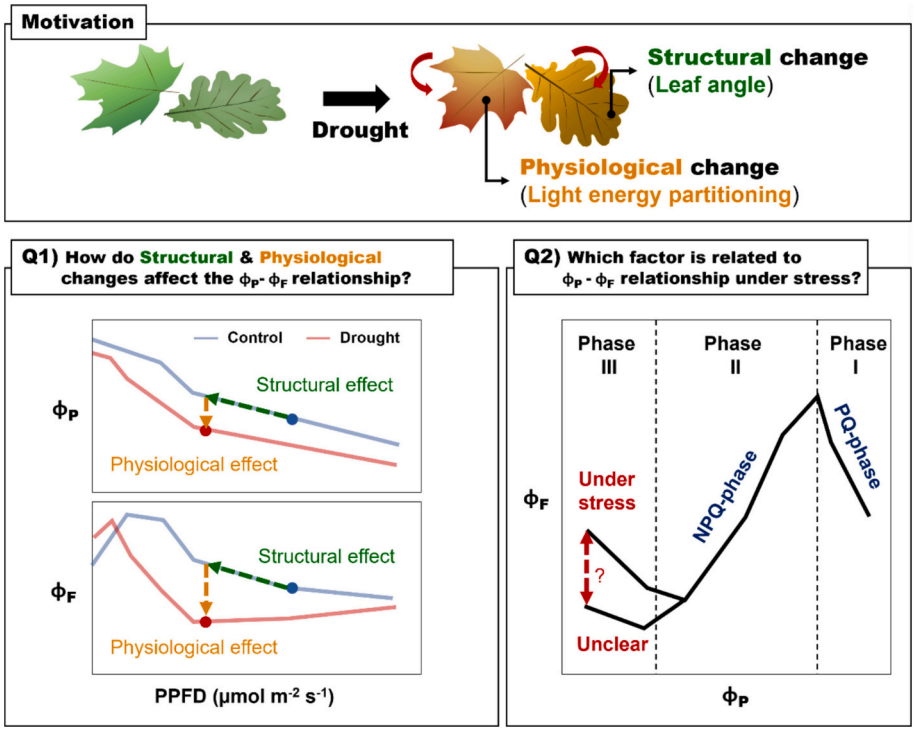


Fig. 1. Conceptual diagram illustrating the study motivation and research questions. We examined leaf angle distribution (LAD) and physiological (light energy partitioning) changes under drought in two deciduous tree species (sugar maple, white oak). Both LAD and physiological changes influence leaf-level light exposure and light-response expression, which together shape the apparent covariation between quantum yields of photochemistry (Φ_P) and fluorescence (Φ_F) observed across leaves. Previous studies have rarely considered these responses simultaneously. Specifically, we address the following questions: (1) How do structural and physiological responses jointly influence the apparent Φ_P-Φ_F covariation? and (2) Which factors influence the Φ_P-Φ_F relationship under drought and high-light conditions? The light-response curves shown in the figure are measured data.

change in PQ.

The Φ_P-Φ_F relationship during drought stress under high light intensity (Phase III, when NPQ is saturated, in Fig. 1) remains challenging to characterize. In this phase, Φ_P is expected to decrease because stomatal closure triggers proton non-consumption mechanisms, and the interplay of these regulatory processes alters the distribution of light energy (Kanazawa and Kramer, 2002; Schreiber and Neubauer, 1990; Flexas et al., 2002a, 2002b). A considerable portion of this energy is dissipated by NPQ before reaching the reaction center, leading to a reduction in PQ (Murchie et al., 1999; Kanazawa and Kramer, 2002; Medrano et al., 2002; Pescador-Dionisio et al., 2025). However, the response of Φ_F to drought stress, particularly in Phase III, is still unclear due to limited and sometimes contradictory findings. For example, van der Tol et al. (2014) reported that Φ_F may increase with rising PAR in leaves previously exposed to drought stress. Other studies have shown that Φ_F decreases alongside Φ_P under drought stress (Ač et al., 2015; Flexas et al., 2002a, 2002b; Ounis et al., 2001). Conversely, drought-driven reductions in Φ_F are not consistently observed at the leaf scale (Chen et al., 2019; Helm et al., 2020; Marrs et al., 2020), and in some cases, Φ_F increases under high light during drought periods (Hwang et al., 2023; Wieneke et al., 2018). This variability in Φ_F responses may be influenced by a range of environmental and biological factors. Therefore, a key question remains: which variables play a key role in the Φ_P-Φ_F relationship under drought and high light intensity when NPQ is saturated?

Investigating the LAD and physiological responses under drought is therefore useful for understanding a more comprehensive picture of light energy partitioning, particularly the relationship between Φ_P and Φ_F (Magney et al., 2020; Porcar-Castell et al., 2014). Specifically, while previous studies have reported that canopy-level SIF reliably estimates photosynthesis under drought stress, why is the relationship between leaf-level Φ_P and Φ_F often inconsistent? We hypothesize that the apparent linear covariation between ΦP and ΦF observed across leaves emerges not only from physiological changes in the intrinsic ΦP–ΦF relationship, but also from drought-driven leaf angle adjustment that redistributes leaves among operating light regimes where ΦP and ΦF exhibit more similar cross-leaf variation. Specifically, we addressed the following three questions:

(1) How do structural (LAD) and physiological variables (Φ_P and Φ_F) change under drought stress conditions?
(2) How do drought-induced changes in LAD redistribute leaves across operating light regimes, and how does this redistribution influence the apparent Φ_P-Φ_F covariation across leaves?
(3) Which physiological variables help explain Φ_F responses under drought and high-light conditions?

To answer the questions, we applied both classical techniques (pulse amplitude modulation (PAM) fluorometer) for monitoring light energy partitioning and newer remote sensing techniques (terrestrial laser scanner, TLS) for quantifying LAD changes in response to drought stress. The answers to these questions are critical to understand why remotely sensed SIF effectively tracks photosynthesis even under drought conditions, as well as to indicate the importance of coordination of structural and physiological aspects for fully understanding light energy partitioning.

2. Materials & methods

2.1. Experimental design

We investigated the effects of drought stress on the two-year-old saplings of two ecologically relevant eastern North America tree species: white oak (*Quercus alba* L.) and sugar maple (*Acer saccharum* Marsh). Two-year-old saplings were selected because one-year-old saplings are highly vulnerable to mortality under experimental conditions, whereas three-year-old saplings are already large to be accommodated within our controlled measurement setup. Individual saplings of both species were planted into plastic pots in high-nutrient compost soils on 1 July 2022 and transferred into walk-in environmental chambers (model PGC-15; Percival, Perry, IA, USA) at Indiana University. We imposed drought treatment on half of the plants (details below) over a two-month period (July to September). All trees received a one-time dose of water-soluble fertilizer with an N:P:K ratio of 20:20:20, and the first measurements were taken on 28 July, 14 days after the initiation of the experiment (i.e., duration of stress or DOS = 14). We collected 3D point cloud data, fluorescence data, and spectral reflectance data every two weeks (Fig. 2).

To mitigate potential differences in plant growth arising from the use of two walk-in environmental chambers of different models, we implemented the following measures. First, although the photosynthetic photon flux densities (PPFD) of the two chambers were different due to difference model of the chambers, we adjusted the distance between the chamber light source and the height of the pots to maintain similar light conditions constantly at the plant canopy. Second, every 7 days, we rotated the saplings both within and among chambers while maintaining the corresponding experimental conditions of each treatment group (Figs. S1 and 2). This procedure was essential to ensure that any differences in plant growth, physiology, or structure were driven by experimentally controlled water availability rather than by potential differences in chamber design or function ("chamber effects"; Porter et al., 2015). Light energy from light emitting diode were evenly distributed throughout the environment chamber (i.e., diffuse light conditions), with a peak value of 650 μmol m^{-2} s^{-1} occurring around noon. This is the maximum PPFD possible within the growth chambers, and while it is less than the PPFD experienced by mature Eastern US oak and maple species, it is comparable to what would be experienced by trees growing in the understory.

To create drought stress, we controlled the water supply in the soil and air (Table 1 and Fig. S1). We created an air drought condition by decreasing humidity and increasing the temperature. The air temperature and vapor pressure deficit (VPD) were initially set to 27.5 °C and 1.2 kPa at DOS 0. For the drought chamber, these conditions were gradually increased, reaching 31.5 °C and 2.4 kPa by DOS 14 and maintained through DOS 28, and further increased to 35.5 °C and 3.5 kPa by DOS 42 and maintained through DOS 56. Soil water supply was manipulated by selective irrigation. Irrigation schedules were based on daily manual measurements of volumetric water content (VWC) at a 20 cm depth using the HydroSense II (Campbell Scientific, Logan, UT, USA). For the control pots, regular irrigations were imposed to maintain abundant soil moisture. In the drought stress pots, irrigation was gradually reduced. At DOS 0, there was no difference in VWC between the control and experimental pots. However, the difference in VWC was around 20% at DOS 14 and DOS 28, and around 40% at DOS 42 and DOS 56. Throughout the experiment, VPD was monitored with temperature and relative humidity probes (HMP-45CL; Campbell Scientific, Logan, UT, USA). VWC was continuously monitored with time domain reflectometry sensors (CS616; Campbell Scientific, Logan, UT, USA and TEROS 12; METER Group Inc., Pullman, WA, USA).

2.2. Leaf angle measurement

We calculated leaf angles by using the difference between the zenith angle and the normal vector in the 3D point clouds collected by TLS (Li et al., 2023; Stovall et al., 2021; Vicari et al., 2019), which is more convenient for calculating numerous leaf angles to traditional methods such as using a handheld inclinometer or image-based approach (Yang et al., 2023; Kattenborn et al., 2022). To acquire information on the leaf inclination angle from 3D point cloud data, we used a Leica RTC360 ultra-fast 3D laser scanner (RTC360, Leica Geosystems., Switzerland). Leica RTC can collect 3D point cloud data with 360-degree horizontal and 300-degree vertical coverage. To collect high-quality 3D point cloud

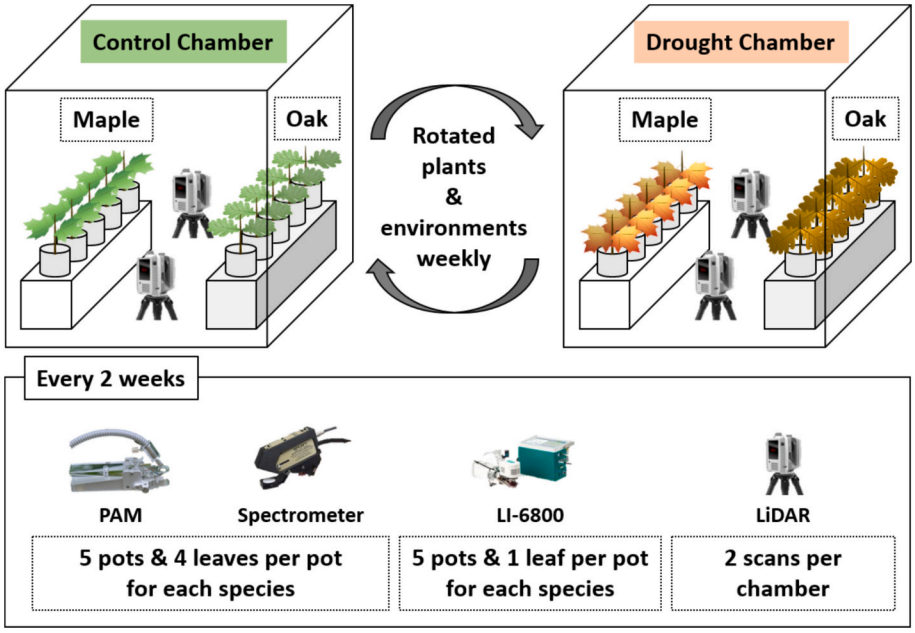


Fig. 2. Schematic figure of experimental design. Detailed methodology for each sensor is explained in the main text. We rotated plants and chamber environments weekly to minimize the chamber effect.

Table 1
Summary of the experimental design parameters applied during the drought experiment. The table lists air temperature, vapor pressure deficit (VPD), and volumetric water content (VWC) measured over the duration of stress (see continuous dataset in Fig. S1). Photosynthetically active radiation (PAR) was maintained at 650 μmol m^{-2} s^{-1} across all treatments and throughout the duration of stress.

	Treatment	Duration of stress (DOS)				
		DOS 0	DOS 14	DOS 28	DOS 42	DOS 56
Air temperature (°C)	Control	27.5	27.5	27.5	27.5	27.5
	Experiment	27.5	31.5	31.5	35.5	35.5
VPD (kPa)	Control	1.2	1.2	1.2	1.2	1.2
	Experiment	1.2	2.4	2.4	3.5	3.5
VWC (%)	Control	45.45	40	40	40	34.7
	Experiment	43.5	25.6	12.7	5.7	7.1

data, we set the high scan resolution setting as 3 mm with 0.4 mm noise at 10 m distance. The scanner was mounted on a tripod at 1.5 m height. To minimize the occlusion problem, in which certain parts of an object are not detected depending on the viewing angle (Wei et al., 2014), two scans were performed for each chamber. Each measurement was conducted at around 13:00. We used Cyclone REGISTER 360 software (Cyclone, Leica Geosystems., Switzerland) to co-register the point clouds from different scanning positions. We used a black-and-white reference panel to merge two scans. The overall mean registration error between the merged two scans was less than ±0.1 cm. We also applied a filter to further reduce noise using the open-source software CloudCompare ("Statistical Outlier Removal tool", Version 2.10.2, General Public License). Finally, we manually isolated individual plants, leaves, and woody parts from the merged 3D point cloud scans by visually identifying and segmenting their geometric and spatial characteristics. This detailed segmentation allowed us to accurately calculate the angles of the points located within the leaf surfaces, excluding points from other plant parts.

To compute the leaf inclination angle, we used a plane-fitting algorithm (Vicari et al., 2019). We employed the nearest neighbor algorithm, selecting 10 neighbors for each observed point cloud (following Vicari et al., 2019), to identify clusters forming leaf surfaces. We then calculated the normal vectors of this surface. Additionally, a surface regression analysis was performed on the cluster of point clouds using singular value decomposition. This analysis yielded a set of eigenvalues and eigenvectors for each point cloud. Finally, we computed the leaf angle by calculating the difference between the normal vectors and the zenith angle. In addition, we confirmed that our leaf angle retrieval method had a reasonable range compared to the triangulation of gridded approaches (Stovall et al., 2021). The distribution of leaf angles was similar between the two different methods (Fig. S2).

To quantify leaf angle changes of plants, we averaged the point cloud data from five pots. In addition, we examined the variance of point cloud angles within each leaf to assess and compare the degree of leaf curling between oak and maple, which provided additional insights into species-specific structural responses. We calculated the standard deviation of the point clouds within the isolated leaf. A higher standard deviation indicated greater variation in leaf angle within a single leaf, resulting from a curling effect.

2.3. Fluorescence measurement

To examine the effect of leaf angle on the relationship between Φ_P and Φ_F, we used a portable photosynthesis system (LI-6800, LI-COR Inc., Lincoln, NE, USA) with a fluorometer head to measure the light response of the quantum yields (Helm et al., 2020; Hwang et al., 2023; Marrs et al., 2020). For each species and each drought treatment, we collected five sets of light response curves on fully expanded leaves growing near the top of the samplings. We fixed the carbon dioxide concentration supplied to the leaf chamber at 420 μmol mol^{-1}, flow rate at 600 μmol s^{-1}, and temperature at 25 °C. In the drought treatment groups, the temperature and relative humidity settings in the LI-6800 were selected to match those of the walk-in environmental chambers (see Experimental design section). Specifically, we set the air temperature and relative humidity in the LI-6800 chamber to 33 °C and 40% for DOS 14 and DOS 28, and to 36 °C and 30% for DOS 42 and DOS 56, respectively. In the control plots, we maintained the values set in DOS 14. The light-response curves of chlorophyll fluorescence were measured by setting the PPFD levels to 900, 600, 450, 300, 200, 100, 50, and 0 μmol m^{-2} s^{-1}. We set the maximum PPFD to 900 μmol m^{-2} s^{-1} because the light intensity of the growth chamber was around 650 μmol m^{-2} s^{-1}. The leaf

was illuminated for 15 to 20 min for each light level until the steady-state fluorescence (F_S) showed consistency (dF/dt: slope < 1, during 60 s). When the F_S value was stable, we recorded F_S, maximum fluorescence in the light (F'_M) using the saturation pulse method, and minimum fluorescence in the light (F'_O). We set the saturation pulse (rectangular flash) as 10,000 μmol m^{-2} s^{-1} for 1 s. We used fluorescence data measured with the LI-6800 to examine the effect of leaf angle on the Φ_P-Φ_F relationship (see Eq. 6 for details).

Additional chlorophyll fluorescence data were collected with a PAM-2500 fluorometer (Heinz Walz Gmbh, Germany). Because the LI-6800 measurements required approximately 30 min for leaf acclimation and 30 min for data collection, the number of samples was limited. To increase the sample size and observe temporal changes in Φ_P and Φ_F under equal light conditions without the influence of leaf angle, we additionally used the PAM-2500, positioning the leaves as flat as possible during measurements. We collected 4 samples per tree and measured a total of 5 trees, resulting in 20 leaves each for control and drought-treated trees. To obtain the chlorophyll fluorescence data under similar light intensity and minimize the effect of leaf angle changes over time, the leaf was placed horizontally while measuring chlorophyll fluorescence data. Therefore, we assumed that chlorophyll fluorescence data observed by PAM-2500 were not affected by changes in leaf angle. The settings of the fluorometer instrument were set to default as an intensity of 10 for measuring light and saturation pulse, and 6 for actinic light. The intensity of the number is an arbitrary unit and is determined by software in the sensor. We applied the same setting for all our measurements during the experiment. Under light from the light-emitting diode of the growth chamber, we recorded F_S, F'_O, and F'_M.

We also measured maximum fluorescence in the dark (F_M) and minimum fluorescence in the dark (F_O) from dark-adapted leaves using two sensors, LI-6800 and PAM-2500. Before each Fm measurement, plants were exposed to a low level of PPFD (< 10 μmol m^{-2} s^{-1}, which represents the minimum light intensity in the growth chamber) for more than 20 min. This dark adaptation period follows the recommendation of Xia et al. (2023) that at least 15–30 min of dark adaptation is required to obtain reliable F_M values.In the case of LI-6800, we could not measure fluorescence with dark-adapted leaves on DOS 14. To mitigate potential errors arising from the use of different instruments, we employed consistent sensors to observe the chlorophyll fluorescence of leaves under both light response and dark adaptation conditions.

For both LI-6800 and PAM-2500 measurements, the quantum yields of Φ_F, Φ_P, and Φ_{NPQ} were calculated using theoretical equations reported elsewhere (Genty et al., 1989; Hendrickson et al., 2004; Kim et al., 2021; Murchie and Lawson, 2013). We have provided a comprehensive explanation of the symbols used for light energy partitioning in the supporting information (Table S1).

To calculate the quantum yields of Φ_F, Φ_P, and Φ_{NPQ} we used formulas from previous studies. Φ_F was calculated by following the theoretical equations (Table A1 and (Genty et al., 1989; Hendrickson et al., 2004; Kim et al., 2021)),

$$\Phi_F = \frac{F_S}{(1 + K_{DF})F_M} \tag{1}$$

K_{DF} Indicates the ratio between the constant rate of constitutive heat dissipation and the constant rate of fluorescence emission. Previous studies have assumed values ranging from 9 to 19 (Atherton et al., 2016; Gu et al., 2019; Porcar-Castell et al., 2006; Zaks et al., 2012). Recent experimental work by Liu et al. (2022) demonstrated that K_{DF} can be estimated from dark-adapted maximum fluorescence quantum yield, and that its value varied only slightly (11.0–12.7) under different temperature and CO_2 conditions. Consistent with these findings and prior studies, we assumed a constant K_{DF} of 9 in this study. To evaluate whether our results depend on the Φ_F estimation method, we estimated Φ_F using a mechanistic light response model (Gu et al., 2019) and confirmed that there was no significant difference in our results (Fig. S3). As a result, we emphasize that our key finding remains robust and is not significantly influenced by Φ_F estimation methods.

The Φ_P could be calculated based on (Genty et al., 1989):

$$\Phi_P = 1 - \frac{F_S}{F'_M} \tag{2}$$

Φ_{NPQ} was estimated by using the equation presented by (Hendrickson et al., 2004).

$$\Phi_{NPQ} = \frac{F_S}{F'_M} - \frac{F_S}{F_M} \tag{3}$$

To quantify the response of Φ_F and Φ_P to changes in light intensity from low levels to maximum used (900 μmol m^{-2} s^{-1}), we applied a simple equation as follows:

$$\Delta\Phi_{F\text{ or }P} = \Phi_{F\text{ or }P\text{ at PAR }450} - \Phi_{F\text{ or }P\text{ at PAR }900} \tag{4}$$

We used the observed light response of quantum yields using LI-6800 to calculate the variation of $\Delta\Phi_F$ and $\Delta\Phi_P$. We defined $\Delta\Phi_F$ and $\Delta\Phi_P$ by subtracting the quantum yields observed at PPFD 450 μmol m^{-2} s^{-1} from the quantum yields observed at PPFD 900 μmol m^{-2} s^{-1}. We chose a PPFD value of 450 μmol m^{-2} s^{-1} because it was half of the maximum light level used (900 μmol m^{-2} s^{-1}). $\Delta\ \Phi_F$ and $\Delta\ \Phi_P$ are intended as relative indicators of the sensitivity of fluorescence and photochemical yields to increasing light intensity, rather than as absolute descriptors of physiological status at a single irradiance.

We also estimated the apparent Cyt b_6f conductance by using a conceptual and quantitative model (Johnson and Berry, 2021):

$$\text{Cyt } b_6f \text{ conductance} = \left(\frac{\text{LEF}}{1 - q_L}\right), \text{where} \tag{5}$$

$$\text{LEF} = \text{PPFD} \times \alpha \times \Phi_P \times \frac{PS_{II}}{PS_I}$$

$$q_L = \frac{F'_M - F_S}{F'_M - F'_O} \times \frac{F'_O}{F_S}$$

where LEF is the linear electron flow and is calculated by multiplying PPFD, leaf absorbance (α), Φ_P, and PS_{II}/PS_I. PS_{II}/PS_I indicates the fraction of that total absorbed PPFD that is absorbed by the antennae of PS_{II}. q_L represents the fraction of the 'open' Photosystem II (PS_{II}) reaction center. We defined the q_L based on a lake-model approach that the reaction centers are connected by a shared antenna (Kramer et al., 2004). According to Johnson and Berry (2021), the value of 1 - q_L is a reasonable estimate for the steady-state condition of the donor side of Cyt b_6f. LEF can also be calculated by multiplying the fraction of Cyt b_6f sites occupied by reduced plastoquinone with the rate at which each reduced site turns over. Therefore, we can estimate the apparent Cyt b_6f conductance to LEF by dividing LEF by (1 - q_L), assuming a lake-model approach. We fitted linear regression between LEF and (1 - q_L) and obtained the slope. We observed Φ_P using the LI-6800 fluorometer. PS_{II}/PS_I was assumed as 0.5, indicating that the electron transfer rates through PS_{II} and PS_I are equal (Genty et al., 1989). We estimated α by using observed reflectance, as explained in the next section.

2.4. Leaf trait estimation using observed spectral reflectance

We used a high-resolution spectroradiometer (SVC Spectroradiometer, Spectra Vista Corporation, USA) equipped with a leaf clip (LC-RP PRO, Spectra Vista Corporation, USA) to collect leaf reflectance. The nominal bandwidth of the spectroradiometer is 1.5 nm from 350 to 1000 nm and approximately 3 nm from 1000 to 2500 nm. The full spectral range (350–2500 nm) was recorded, and to quantify reflectance and absorbance, we processed the data after acquisition to extract only the wavelengths within the photosynthetically active radiation range of 400–700 nm, which plants primarily use for photosynthesis. We obtained 20 reflectance spectra from the same set of leaves that were measured using the PAM-2500. We used a low-intensity light source in

the leaf clip because the leaves were burned when we used the strong-intensity setting. To collect high-quality reflectance data, we measured a white reference when we collected around 5 samples.

We estimated leaf absorbance (α) using leaf-level reflectance data via the inversion of the PROSPECT radiative transfer model (Jacquemoud and Baret, 1990). To calculate α, the transmittance data is essential. However, we were unable to measure the transmittance without cutting the leaves, and the number of leaves available was limited. Therefore, we estimated the α using the radiative transfer model (Detailed flow chart in Fig. S4).

2.5. The function of leaf angle, absorbance, and quantum yields

To demonstrate the effect of changes in leaf angle and absorbance on the observed Φ_F -Φ_P relationship, we calculated the leaf-angle-adjusted Φ_F ($\Phi_{F_{angle}}$) and Φ_P ($\Phi_{P_{angle}}$), since leaf-level chlorophyll fluorescence measurements using PAM-2500 assume that the incident light is perpendicular to the leaf surface. Explicitly accounting for leaf angle, given its well-known impact on light interception, provides a more realistic estimation of Φ_F and Φ_P (Fig. 1). Importantly, plants can regulate leaf angle independently of photosynthetic metabolism, providing a mechanism to adjust quantum yields without directly altering the biochemistry of photosynthesis. To explore this, we analyzed leaf angle obtained from TLS together with photosynthetic light-response data measured by the LI-6800 to assess how leaf angle adjustments shift leaf-level absorbed PPFD and thereby influence the apparent Φ_F-Φ_P covariation observed across leaves.

The amount of light absorbed by a leaf is a function of the incoming solar irradiance, the leaf angle, and absorbance (α). We derived a simple equation to calculate the values of Φ_P and Φ_F that correspond to the estimated absorbed PPFD, adjusted by leaf angle and absorbance (Eq. 6). It is important to note that our approach is different from directly multiplying Φ_F or Φ_P with cos(leaf angle).

$\Phi_{F_{angle}}$ and $\Phi_{P_{angle}}$ are observed Φ_F and Φ_P at PPFD ω_{angle}, where (6)

PPFD ω_{angle} is the adjusted incident light interception and equal to
$650\ \mu mol\ m^{-2}\ s^{-1} \times \cos(leaf\ angle)$

$\Phi_{F_{angle\&\alpha}}$ and $\Phi_{P_{angle\&\alpha}}$ are observed Φ_F and Φ_P at PPFD $\omega_{angle\&\alpha}$, where

PPFD $\omega_{angle\&\alpha}$ is the adjusted incident light interception, absorbance and equal to
$650\ \mu mol\ m^{-2}\ s^{-1} \times \cos(leaf\ angle) \times \alpha$

Leaf angle was measured by TLS, α was estimated by using observed reflectance with the radiative transfer model. The light response curves of Φ_F and Φ_P were initially measured separately for each species (maple and oak) and each water treatment (control and drought) using the LI-6800 system. The Φ_F and Φ_P values derived from these light response curves and Eq. 6 were then aggregated by averaging across measurements either within each species or within each water-stress condition, depending on the analytical objective. The amount of light reaching the leaf surface could be estimated as the product of incoming solar irradiance and cos(*leaf angle*) when sunlight strikes the ground perpendicularly (Verhoef, 1984). Therefore, the actual amount of light absorbed by the leaf (PPFD ω) in Eq. 6) can be calculated by multiplying the PPFD of the chamber LED by the cos(*leaf angle*) and α. Because this simple equation represents light intercepted by the upper surface of a leaf, we also verified the results using Soil Canopy Observation of Photosynthesis and Energy fluxes (SCOPE) simulations (Van der Tol et al., 2009, 2014; Yang et al., 2021), which account for radiation incident on both sides of the leaf. The SCOPE-based estimates confirmed that our key message remained unchanged (see Discussion section and Appendix B).

To compute the output quantum yields, we interpolated the observed light response of Φ_F and Φ_P from LI-6800 using the 'interp1' function in MATLAB (The MathWorks Inc., Natick, MA, USA). Then, we chose Φ_F and Φ_P as values that corresponded to the calculated PPFD ω from the interpolated Φ_F and Φ_P. In this way, our approach allowed us to independently assess the effects of LAD on the Φ_F-Φ_P relationship by explicitly separating structural adjustments (leaf angle and absorptance) from physiological responses.

2.6. Statistical analysis

We used a one-way analysis of variance (ANOVA) with the assumption of homogeneity of variance followed by Dunn-Sidák's multiple comparison tests in MATLAB to test whether there was a statistically significant difference between control and experiment trees, or differences over time within the same treatment. We have established that a nonsignificant relationship exists when the *p*-value exceeds 0.05. We use varying numbers of asterisks to indicate significant differences (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). To evaluate the relationship between two variables, the linear ordinary least squares regression analysis was performed, and the coefficient of determination (R^2) was employed to quantify the goodness of fit of the linear regression model.

3. Results

The leaf angles of maple and oak increased (i.e., became steeper) in response to drought (Figs. 3 and S5). Before drought stress, there were smaller differences in leaf angle between the two species and treatments (Figs. 3b and c). Under drought stress, the mean leaf angle of maple increased from 46.5 degrees (at DOS 14) to 62 degrees (at DOS 56). Oak also experienced a drought-driven increase in mean leaf angle from 50.8 degrees (at DOS 14) to 57.1 degrees (at DOS 56). This increase was less than half of that observed in maple (15.5 degrees (33%) for maple and 6.3 degrees (12%) for oak). The control trees also showed the leaf angle changes vertically (7.1 degrees (14%) for maple and 3.8 degrees (7.5%) for oak, in Fig. S6), but these changes were significantly lower than those observed in trees affected by drought stress.

We also found that the variation in leaf angle clearly differed between the species under drought stress. The number of maple leaves with steeper (higher) leaf angles increased as the drought continued (Fig. 3b). We observed a similar trend in oak, except that leaf angle frequency drops after 60.2 degrees (Fig. 3c). Furthermore, there was a notable difference in leaf morphological responses to drought stress between maple and oak trees. Specifically, the maples exhibited leaf shrinking and curling, while the oaks tended to adjust the planar orientation of the entire leaf in response to drought conditions (Fig. 3a). This pattern was clearly demonstrated by the variance in leaf angle within each leaf. Under drought stress conditions, the standard deviation of points clouds within one single leaf was 19.6 degrees for maple and 10.8 degrees for oak (Fig. S6).

Both Φ_P and Φ_F observed with the PAM-2500 significantly decreased in maple and oak in response to drought conditions. In the PAM-2500 measurements, we positioned the leaves flat so that the light intensity on the leaf surface remained relatively constant over time. The magnitude of decrease in Φ_P (38%) was nearly identical for maple (Fig. 4a) and

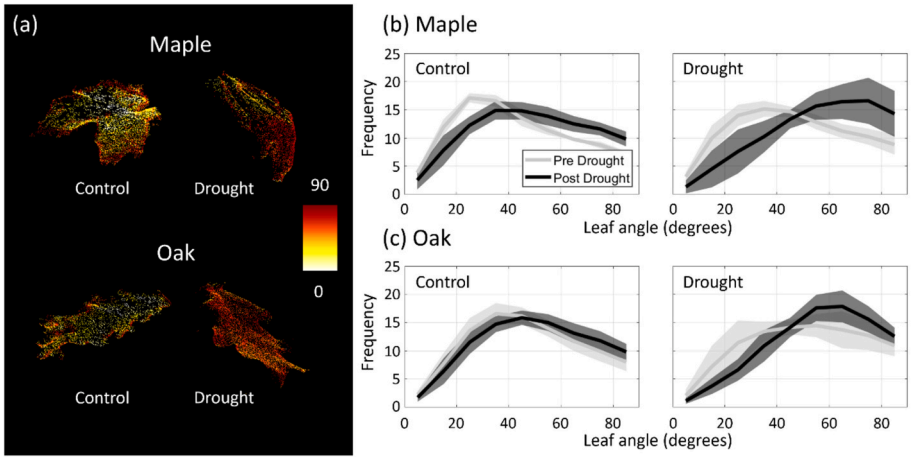


Fig. 3. (a) The high-resolution estimation of point cloud angles from the terrestrial laser scanner and leaf inclination angle distribution in control and drought conditions for (b) maple and (c) oak. The colormap in (a) indicates the angle of point clouds. In (b) and (c), the grey lines indicate the averaged frequency of angle distribution before drought and the black lines represent the averaged frequency after drought. 0 degrees is horizontal, and 90 degrees is vertical. Observed data at the duration of stress (DOS) 14 and DOS 56 were used for pre-drought and post-drought results. The shaded area represents 95% confidence intervals. For the time-series of mean and variance of leaf angles, see Fig. S5.

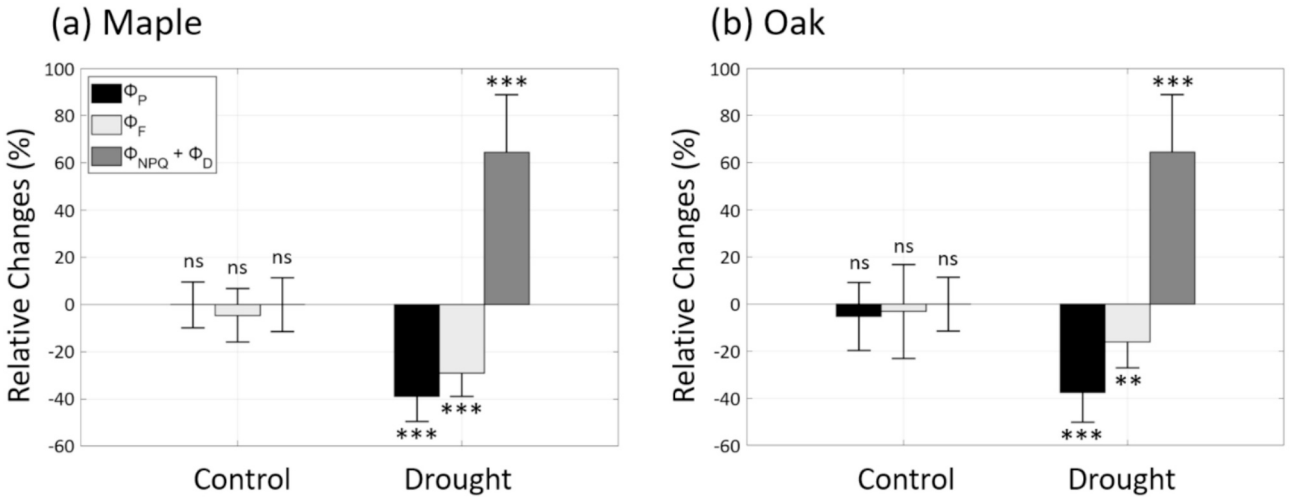


Fig. 4. The drought response of quantum yield of photochemistry (Φ_P, black), chlorophyll fluorescence (Φ_F, white), and non-photochemical quenching and heat dissipation ($\Phi_{NPQ} + \Phi_D$, grey) in (a) maple and (b) oak over time. Data were collected using the PAM-2500, with leaves positioned flat to keep the light intensity on the leaf surface relatively constant throughout the measurements. The error bar represents 95% confidence intervals. The statistically significant differences between the duration of stress (DOS) 14 and DOS 56 were calculated by analysis of variance (ANOVA, for the beginning of the experiment, the sample size of treatments = 19, controls = 20. For the end of the experiment, the sample size of treatments = 11, n control = 13, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). For the time series of Φ_P, Φ_F, and Φ_{NPQ}, see Fig. S7.

oak (Fig. 4b), whereas the decrease in Φ_F under drought was greater in maple (29%) than oak (16%). In the control group, there were no significant changes in Φ_P, Φ_F and Φ_{NPQ+D} over time (Figs. 4 and S7).

When we assessed the covariation between Φ_P and Φ_F across all the leaves, we found that accounting for variation in leaf angle and absorbance made the apparent covariation between Φ_P and Φ_F across leaves more linear in maple and oak (Fig. 5). When no leaf angle corrections were included in Eq. 6, there was no significant relationship between Φ_P and Φ_F (Fig. 5a). However, when leaf angle effects were included, the relationship between Φ_P and Φ_F was positively linear and significant (Fig. 5b, $R^2 = 0.41$, $p < 0.05$). The apparent cross-leaf Φ_P-Φ_F covariation became more linear, even when looking at each species individually (oak: from $R^2 = 0.43$ to 0.75, maple: from $R^2 = 0.34$ to $R^2 = 0.88$). Notably, additionally accounting for absorbance further increased the linearity of the apparent Φ_P-Φ_F covariation across leaves (Fig. 5c, $R^2 = 0.60$, $p < 0.001$, and $R^2 = 0.83$ for oak and $R^2 = 0.89$ for maple, separately). Although the leaf angle did not significantly change in control trees, we also found that the variations in leaf angle among individual leaves could make the apparent Φ_P-Φ_F covariation across leaves more linear (Fig. S8).

The Φ_P-Φ_F relationship varied between the two species (Fig. 6). Φ_P in maple first increases with Φ_F and has a negative relationship with Φ_F after the value of Φ_P reaches approximately 0.65 (Fig. 6a). On the other hand, the Φ_P-Φ_F relationship of oak showed a wavy pattern that differed from that of maple (Fig. 6b). For example, the Φ_P-Φ_F relationship was negative when Φ_P had lower than 0.25 and higher than 0.6 under drought stress, but showed a positive relationship when Φ_P was from around 0.25 to 0.6. The wavy pattern of the Φ_P-Φ_F relationship was clearly observed when normalized quantum yields were used (Fig. 6c and d). Although the Φ_P-Φ_F relationship of the two species showed different patterns, both species experienced a decrease in their maximum and minimum value of Φ_P value under drought stress

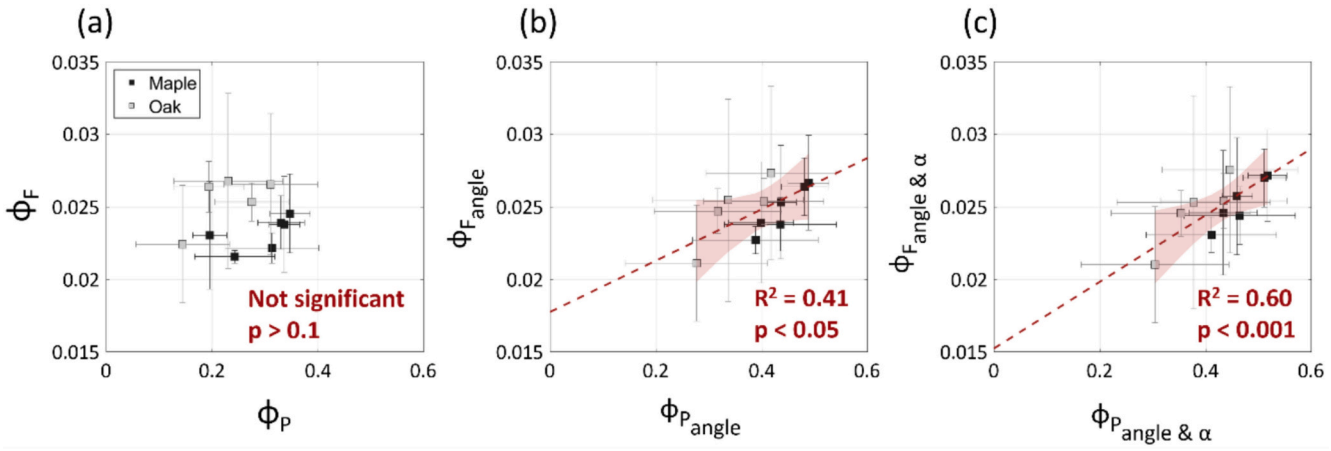


Fig. 5. (a) The apparent covariation between quantum yields of photochemistry (Φ_P) and chlorophyll fluorescence (Φ_F) across all the leaves, at 650 μmol m^{-2} s^{-1}, (b) the correlation between angle-adjusted quantum yield of photochemistry ($\Phi_{P_{angle}}$) and chlorophyll fluorescence ($\Phi_{F_{angle}}$); here, these values were estimated from measured light-response curves, using a value of PPFD that match what the leaves would intercept at a given leaf angle and light intensity, and (c) the correlation between angle-absorbance-adjusted quantum yield of photochemistry ($\Phi_{P_{angle\&\alpha}}$) and chlorophyll fluorescence ($\Phi_{F_{angle\&\alpha}}$), again calculated as corresponding values of Φ_P or Φ_F from light response curves at the PPFD when considering the leaf angle and absorbance. To create scatter plots, we used the averaged light response curves of Φ_P and Φ_F measured with LI-6800 (not the PAM-2500) and observed leaf angle using a 3D laser scanner from five pots. See the Method section for details. Data from the duration of stress (DOS) 28 to 56 were used. To increase the sample size, we combined the data from control and drought-treated trees. For separate relationships between control and drought-treated trees, see Fig. S8. Each data point with an error bar comes from leaves of the same pot, and it indicates the mean of light response data measured at each DOS with ±95% confidence intervals. The dashed red line indicates the linear regression line, and red shaded areas represent 95% confidence intervals. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

conditions (Fig. 6).

We found a strong positive linear relationship between $\Delta\Phi_P$ and $\Delta\Phi_F$ (Eq. 4), regardless of the species and treatments (Fig. 7a, $R^2 = 0.58$, $p < 0.001$). Furthermore, we found that $\Delta\Phi_F$ positively correlates with Cyt b_6f conductance (Fig. 7b, $R^2 = 0.82$, $p < 0.001$).

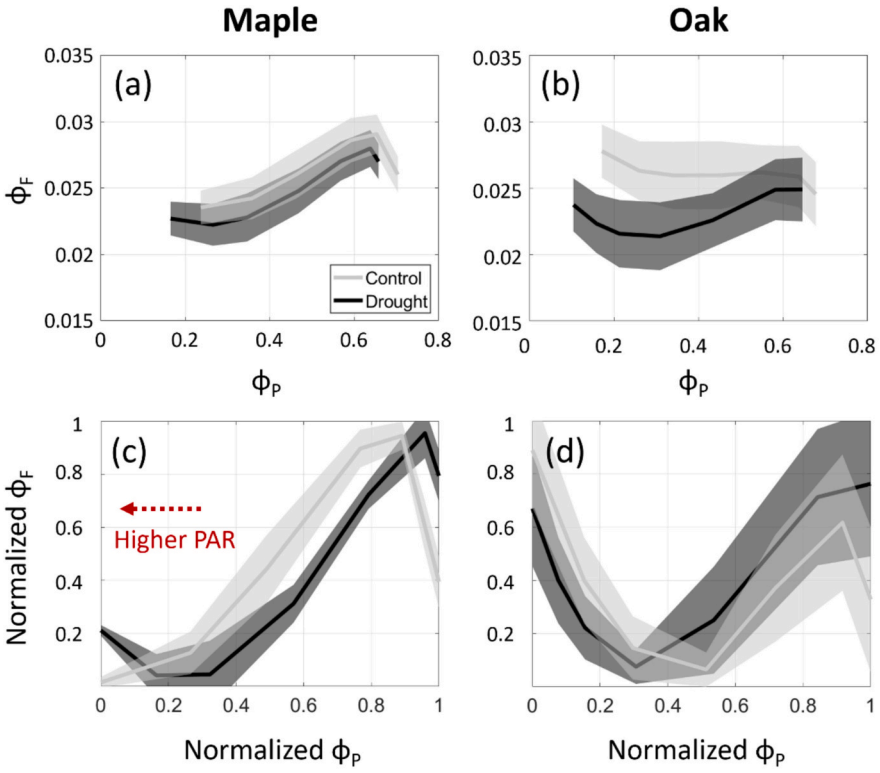


Fig. 6. Relationship between quantum yield of photochemistry (Φ_P) and quantum yield of chlorophyll fluorescence (Φ_F) in (a) maple and (b) oak, and between normalized Φ_P and Φ_F in (c) maple and (d) oak. The white color indicates the relationship between control and experiment plants and the black color represents the relationship under drought stress. Observed data from the duration of stress (DOS) 28 to 56 were used. We averaged the Φ_P-Φ_F relationship from light response curve observed by LI-6800 for control and drought-treated trees. The shaded area represents 95% confidence intervals.

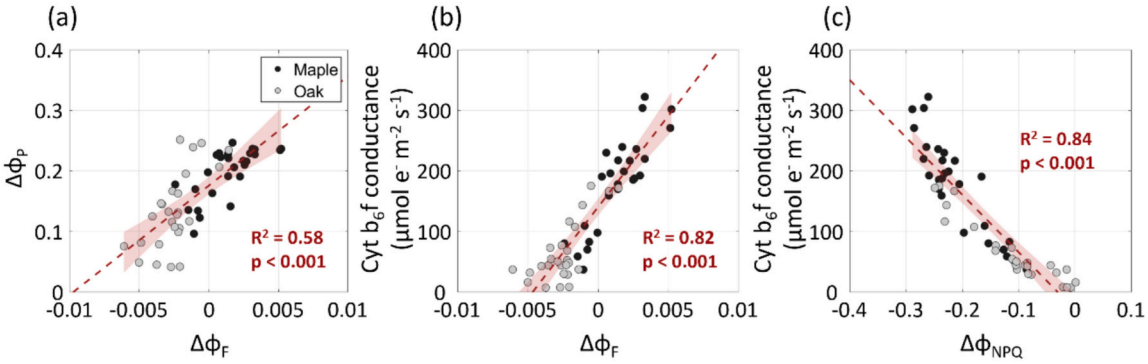


Fig. 7. Relationship between changes in quantum yield of chlorophyll fluorescence ($\Delta\Phi_F$) and (a) photochemistry ($\Delta\Phi_P$), (b) Cyt b_6f conductance (μmol e^- m^{-2} s^{-1}), (c) and relationship between Cyt b_6f conductance and changes in quantum yield of non-photochemical quenching ($\Delta\Phi_{NPQ}$). The black and grey dots indicate maple and oak, respectively. Observed data from the duration of stress (DOS) 28 to 56 were used. The data points include those from both control and droughted trees. The dashed red line indicates a linear regression line, and the red shaded areas represent 95% confidence intervals. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4. Discussion

This study aims to address the following question: While numerous studies report that canopy-level SIF reliably estimates photosynthesis, why is the relationship between leaf-level Φ_P and Φ_F often inconsistent under drought stress conditions? We hypothesize that LAD plays a critical role in shaping the apparent Φ_P-Φ_F relationship at the canopy scale. To test this, we quantified LAD and light energy partitioning parameters to investigate the underlying mechanisms of the Φ_P-Φ_F relationship under drought stress. For both maple and oak, leaf angles increased in response to drought; however, maple leaves exhibited greater curling, while oak tended to adjust leaf orientation more planarly, resulting in notably distinct morphological responses. The apparent Φ_P-Φ_F relationship across leaves became more linear when variation in leaf angle and absorbance was accounted for, although maple and oak exhibited different magnitudes of decrease in Φ_P and Φ_F under drought conditions. In the following section, we discuss the implication of these results in relation to the specific questions introduced in the Introduction, focusing on the structural and physiological aspects and the impact of LAD changes on the Φ_P-Φ_F relationship. We also highlight that an understanding of both structural and physiological mechanisms at the leaf level is essential for accurately interpreting remote sensing signals at the canopy scale.

4.1. What are the structural changes under drought stress?

The leaf angles of both oak and maple increased under drought stress (Figs. 3 and S5) supporting the notion that plant leaves become more vertical to reduce light absorption and leaf surface temperature (Yang et al., 2023). Interestingly, the average leaf angle of maple increased approximately twice as much as that of oak (Figs. 3 and S5). Our estimated equivalent water thickness (EWT) using spectral data also supported this, as the EWT of maple showed a significant decrease in drought-treated trees, while the decrease in EWT for oak was not as pronounced (Fig. S9). Our results differ from the findings of Casey (2020), who used image analysis to show that black oak, but not white oak, increased leaf angle more than sugar maple in response to reduced soil moisture conditions. The "shrinking (i.e., curling leaf surfaces)" of maple leaves may explain the difference (Fig. 3). Although the angle of the main leaf veins in maple varied less than that of oak (Fig. 3a), conspicuous curling of both sides of the main leaf vein of the maple occurred. In the curled area, the leaf angle significantly increased (Fig. S6). On the other hand, oak did not shrink until the leaves turned brown due to desiccation. The observation of leaf shrinkage may be difficult for digital image-based analysis to capture, because such methods typically have limited spatial resolution and rely on 2D projections, which may not detect subtle three-dimensional changes in leaf curvature or inclination.

4.2. How does leaf angle adjustment shape apparent Φ_P-Φ_F covariation?

Changes in leaf angle (Fig. 3) and absorbance can impact the absorbed PPFD, ultimately enhancing the relationship between Φ_P and Φ_F (Question 1 in Fig. 1). We found that the Φ_P-Φ_F relationship became more interpretable when integrating leaf angle dynamics (From not significant to $R^2 = 0.41$ with $p < 0.05$, Fig. 5b). It is important to emphasize that LAD does not alter the intrinsic Φ_P-Φ_F relationship at the level of a single leaf under fixed irradiance. Rather, LAD influences the operating light environment across leaves, thereby shaping the Φ_P-Φ_F covariation observed under heterogeneous light conditions. Regardless of drought stress, leaf angle changes were associated with a higher correlation in the Φ_P-Φ_F relationship (Fig. S8). Leaf angle variation makes the apparent cross-leaf Φ_P-Φ_F covariation more linear for the following reasons. First, the change in leaf angle shifts the operating light regime experienced by the leaf by reducing leaf-intercepted irradiance, which in turn leads to an increase in Φ_P (Fig. S10). Conversely, Φ_F shows less variation in response to change in light intensity in Phase II (Figs. 6 and S10). For instance, when the incident PPFD was decreased from 800 to 200 μmol m^{-2} s^{-1}, Φ_P increased by approximately 230%, whereas Φ_F only experienced an 8% change (Fig. S10). Therefore, when leaf angle dynamics are considered, the apparent decline in Φ_P observed in Fig. 4 would be smaller, because a reduced light load on the leaf surface mitigates the drought-induced decrease in Φ_P. As a result, the relative changes in Φ_P become more comparable to those of Φ_F. Second, leaf angle change reduces the amount of intercepted light, thereby limiting the possibility of leaves reaching Phase III (Fig. 1), during which a negative Φ_P-Φ_F relationship could be observed. For example, although oak trees show a negative Φ_P-Φ_F relationship under strong light conditions (Figs. 6 and S10), this reduced the likelihood of a negative relationship under drought stress conditions because the vertical leaf angle reduces the possibility of intense light absorbance. Consequently, Φ_P and Φ_F can show a positive linear relationship (Fig. 5). Although the changes in absorbance were not clearly observed (Figs. S11 and S12), they could still affect the amount of absorbed light, thereby contributing to the apparent Φ_P-Φ_F covariation (Fig. 5c). Additionally, we confirmed that the response of Φ_P and Φ_F to drought stress conditions showed a similar decrease by assuming a constant incoming solar energy of 100% and taking into account variations in leaf angle, absorbance, and light energy partition (Fig. 8). This overview of the flow of light energy suggests a potential explanation for the change in Φ_P-Φ_F relationship due to leaf angle change.

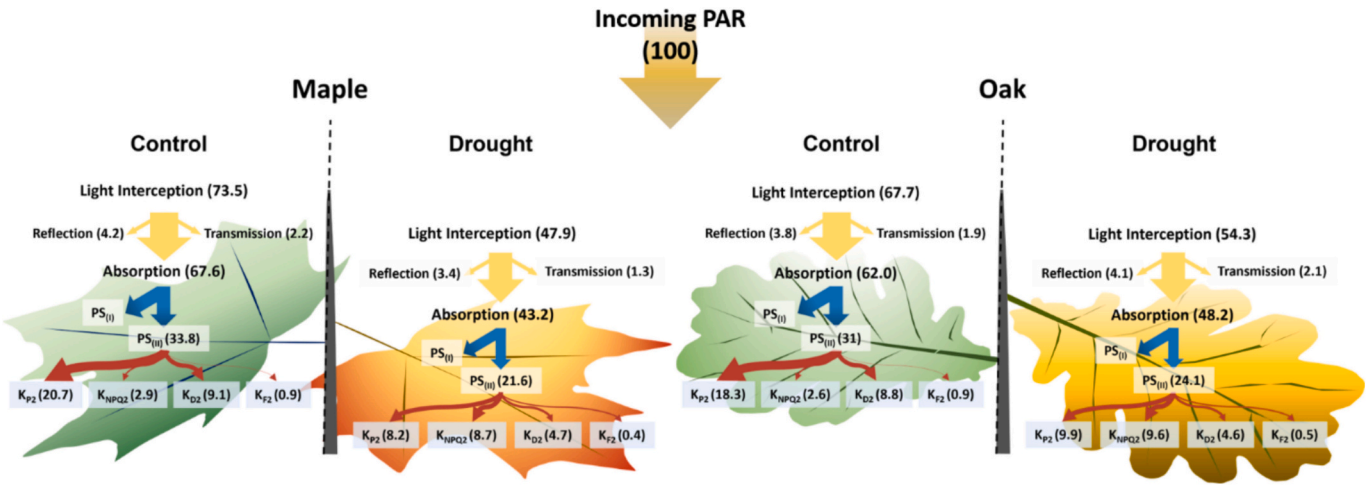


Fig. 8. Overview of the flow of light energy within maple and oak leaves in the control and experiment groups. This is a schematic diagram showing how much light energy corresponds to each stage based on in-situ observation values, assuming the incoming PAR is 100%. The amount of light intercepted by the leaf changes as the leaf angle changes, and the corresponding quantum yields are shown to vary based on changes in light energy partitioning within the leaf. The average ratio of photosystem II (PS_{II}) reaction centers to photosystem I (PS_I) reaction centers was assumed as 0.5. We calculated absorbance (α) by subtracting the observed reflectance and estimated transmittance from 1. We used the photosynthetically active radiation region (400–700 nm) to quantify reflectance and α. The combined value of arrows of the same color indicates the total light energy of each step. The rate constants for PS_{II} photochemistry, non-photochemical quenching, constitutive heat loss in the antennae, and fluorescence are denoted as K_{P2}, K_{NPQ2}, K_{D2}, and K_{F2}, respectively.

4.3. How does light energy partitioning change under drought stress?

Φ_P decreased more than Φ_F under drought stress conditions regardless of the species (Fig. 4). We confirmed that as DOS increased, Φ_{NPQ} increased and Φ_P decreased correspondingly (Fig. S7) in a similar manner observed by previous studies (Flexas et al., 2002a, 2002b; Helm et al., 2020; Ounis et al., 2001). We also found that Φ_F also decreased over time, but to a more marginal extent than Φ_P (Helm et al., 2020; Marrs et al., 2020). Based on these findings, during drought, a significant portion of the light energy is redirected towards NPQ, decreasing in Φ_P and potentially smaller reduction in Φ_F (Figs. 4 and S7).

The degree of change in Φ_F in response to drought could be different depending on the species. Φ_F in maple decreased by 29%, while Φ_F of oak decreased by 16% (Figs. 4a and b). The different response of Φ_F in oaks and maples to drought stress is also evident in the distinct patterns of the Φ_P-Φ_F relationship (Figs. 6c and d). Furthermore, under high light conditions, Φ_F in oak increased again, whereas Φ_F in maple did not (Figs. 6 and S10). Consequently, most of the $\Delta\Phi_F$ values in oak showed positive values, while most of $\Delta\Phi_F$ in maple were negative (Fig. S13). Since Φ_F in oak tends to re-increase under extreme stress conditions (e.g., Phase III and Question 2 in Fig. 1), the degree of Φ_F reduction in oak during drought could be less than that in maple. Then, why did oak Φ_F increase under high light (i.e., higher $\Delta\Phi_F$) compared to maple?

$\Delta\Phi_P$ and Cyt b_6f conductance were significantly correlated with $\Delta\Phi_F$ (Fig. 7). As light intensity increases, Φ_P decreases while Φ_{NPQ} increases (Fig. S10). Once electron flow for photochemistry and NPQ reaches saturation, the remaining energy could be released as additional fluorescence and constitutive heat, resulting in an increase in Φ_F. For this reason, $\Delta\Phi_P$ and $\Delta\Phi_{NPQ}$ could be strongly correlated with $\Delta\Phi_F$ (Figs. 7 and S14). The light response of Φ_P and Φ_F might be influenced by the carotenoid: chlorophyll ratio, Cyt b_6f conductance, and V_{cmax} (Johnson and Berry, 2021; Wong and Gamon, 2015; Zhang et al., 2016). We confirmed that $\Delta\Phi_F$ and carotenoid: chlorophyll ratio estimated using spectral reflectance showed a weak correlation (Fig. S15a). Additionally, the correlation between $\Delta\Phi_F$ and nitrogen content was weak, which exhibited a relatively high correlation with V_{cmax} (Figs. S15b and S16b). However, Cyt b_6f conductance explained 82% of the variations in $\Delta\Phi_F$ (Fig. 7b). Cyt b_6f conductance is crucial for regulating the linear flow of electrons through the plastoquinone pool and the concomitant proton translocation into the thylakoid lumen (Johnson and Berry, 2021). This implies that the lower Cyt b_6f conductance limited electron flow from the plastoquinone/plastoquinol pool, the closure of PS_{II} reaction centers, and subsequently lead to greater energy dissipation via fluorescence and constitutive heat. In other words, the low Cyt b_6f conductance causes the saturation of electron flow for PQ and NPQ, resulting in a subsequent increase in Φ_F under intense light conditions (Fig. 7).

4.4. Limitations

Several limitations of the study should be considered when interpreting the results. First, estimated α and leaf traits from leaf spectra may introduce uncertainties. We derived transmittance and leaf traits using spectral reflectance and a radiative transfer model to avoid destructive sampling of leaves as required by the instrument setup. However, leaf traits and inversed transmittance estimated solely from reflectance may introduce bias. Although it is important to note that the model inversion results have their uncertainties compared to in-situ observations, PROSPECT has been validated with leaf samples from various species (Féret et al., 2021; Spafford et al., 2021). In addition, some leaves turned brown before abscission, and these were excluded from our analysis because reasonable chlorophyll fluorescence data could not be obtained from them. As a result, in our dataset the shape of spectral reflectance did not change significantly during the experiment. Second, while TLS enables the calculation of leaf angle for individual leaves, it was challenging to determine which leaf was measured by the LI-6800 in the 3D point cloud. Consequently, we used the averaged leaf angle of the whole plants for Eq. 6 which may increase the uncertainty (Fig. 5). Applying both structural and physiological changes to the same leaf could potentially yield results with a smaller margin of error. Third, the simple equation we used to estimate the intensity of direct light incident on the leaf is insufficient to represent actual environmental conditions. We assumed that only direct light from the zenith direction reached the leaf and calculated intercepted light using a simplified equation (Eq. 6). Nevertheless, using the SCOPE simulations, a widely used radiative transfer model for simulating SIF under realistic light conditions (Van der Tol et al., 2009, 2014; Yang et al., 2021), we confirmed that changes in leaf angle reduce absorbed light and increase the apparent linearity of Φ_P-Φ_F covariation across leaves, consistent with our key findings (see Appendix B). Fourth, further research is needed to better explain the photosynthesis-SIF relationship by

examining additional physiological and structural processes (Hwang et al., 2023; Xu et al., 2021). Although we identified potential reasons for the linearity of the Φ_P-Φ_F relationship, processes involved in carbon metabolism (e.g., photorespiration) *directly* modulate gas exchange (and thus photosynthesis) but not fluorescence. At the canopy scale, factors such as leaf area, leaf angle, and clumping are crucial in determining the total amount of intercepted light and escaped fluorescence (Dechant et al., 2020; Fournier et al., 2012; Li et al., 2022; Yang et al., 2020; Yang et al., 2023; Zeng et al., 2019) as well as gas exchange. In fact, the average number of leaves and point clouds were significantly decreased under drought (Table S1). Nonetheless, studies that simultaneously observed leaf-level light energy partitioning, gas exchange, and canopy-level structural information remain scarce. Therefore, additional research is needed to clarify SIF-photosynthesis relationships from the leaf to the landscape level.

4.5. Recommendations

Some research questions have arisen based on our results, and further research needs to be conducted to answer them. We found significant differences in $\Delta\Phi_F$, $\Delta\Phi_P$, and Cyt b_6f conductance between species. Additionally, the slope of the Φ_P-Φ_F relationship in oak control trees was clearly different from that of others. Although we conducted further analysis on pigment pool changes (Figs. S17 and S18) and statistical analysis in quantum yields and Cyt b_6f conductance between control and drought-treated trees (Fig. S19), we were unable to pinpoint the exact reasons due to the limited number of samples. Our hypothesis suggests that there might be different mechanisms influencing light energy partitioning in oak and maple. For example, oak and maple have contrasting water-use strategies. The hydraulic function may be an important regulator of leaf angle change, and this will be the focus of future work. (Benson et al., 2022; Yi et al., 2020). In addition, photosynthetic capacities and responses to illumination conditions may vary with plant age (Hinckley et al., 1978; Liu et al., 2021) although light intensity in the growth chamber exceeds the saturation point (Fig. S20). Therefore, further research is recommended on natural forests.

To accurately understand the mechanism of reduction in SIF under drought stress conditions, it is recommended to observe both the structural and physiological variables (Sun et al., 2023; Yang et al., 2023; Jablonski et al., 2025). In this paper, we take into account the structural changes, including leaf angle, to reveal the actual Φ_P-Φ_F relationship. Leaf-level measurements that do not consider leaf angles, which directly impact the light intensity at the leaf level, may inadvertently overlook the impact of light absorption on Φ_P and Φ_F. For example, some studies observed either an increase or no significant difference in Φ_F compared to Φ_P under drought stress with high light conditions (Hwang et al., 2023; Marrs et al., 2020). It could be possible because some species (e.g., the oak species in our study) can have the characteristic of re-increasing Φ_F at high light level. Therefore, to understand the response of Φ_F to drought stress conditions, it is necessary to recognize that the response of Φ_F can be changed depending on the value of PAR (Chen et al., 2019). Accordingly, leaf-level fluorescence measurements should be conducted at the natural leaf position (inclination and azimuth angles) to reflect the actual light environment experienced by the leaf. Furthermore, we observed variations in the Φ_P-Φ_F relationship among species (Fig. 6). This different Φ_P-Φ_F relationship was found to be associated with the light response of Φ_F and was closely linked to Cyt b_6f conductance (Fig. 7). This finding can be used to improve modeling works for SIF simulation. For example, the SCOPE relies on empirical equations for estimating the changes in Φ_F under varying light intensities (Van der Tol et al., 2009, 2014).

Beyond the leaf scale, canopy structural factors strongly influence canopy-level SIF observed by satellite or tower-based measurements. To scale our chamber-based results to natural canopies, it is therefore essential to integrate LAD with other structural parameters, including leaf area index, clumping index, and three-dimensional canopy architecture. Canopy structure governs the escape probability of fluorescence photons, as SIF emitted from individual leaves undergoes multiple scattering events before exiting the canopy (Fournier et al., 2012; Li et al., 2022; Yang and van der Tol, 2018). In particular, LAD modulates leaf-level light interception: vertically oriented leaves reduce total absorbed radiation (Figs. 3 and 8), and drought-induced reductions in leaf number further decrease canopy light absorption (Table 1), collectively leading to substantial changes in canopy-level SIF (Hwang et al., 2023; Xu et al., 2021). These results highlight that structural adjustments under stress conditions exert a significant control on SIF signals beyond physiological regulation alone. Recent advances in satellite-derived LAD and leaf area index products (Li et al., 2022; Wan et al., 2024; Chen et al., 2025), together with terrestrial LiDAR and drone-based three-dimensional canopy reconstructions (Jablonski et al., 2025), provide new opportunities to extend our approach to more complex and realistic ecosystems. We therefore suggest that future studies combine structural proxies with physiological measurements under field conditions to improve LAD-informed SIF modeling frameworks. More broadly, linking canopy-scale remote sensing observations to leaf-level processes requires explicit consideration of canopy structure and the actual light environment. Integrating our findings with satellite- or tower-based SIF measurements will ultimately enable scaling from controlled experiments to regional and global assessments of plant stress and ecosystem function.

5. Conclusion

SIF has been widely used as a proxy for understanding global carbon uptake in vegetation. However, studies that simultaneously observe structural and physiological changes to clarify how SIF effectively tracks photosynthesis, even under drought stress, are limited. We found that changes in leaf angle could be related to the linear Φ_P-Φ_F relationship, although the degree of physiological variation in Φ_P-Φ_F differs across light and drought stress conditions. Additionally, we observed that Cyt b_6f conductance is related to PQ and NPQ saturation, resulting in increased Φ_F under high-light conditions, which determined the Φ_P-Φ_F relationship. These findings highlight the importance of jointly considering physiological and structural changes as this approach potentially explains why SIF is an indicator of photosynthesis under drought stress. We suggest that observing either physiological or structural changes alone may lead to an incomplete understanding of the SIF-photosynthesis relationship under normal or drought conditions.

CRediT authorship contribution statement

Jongmin Kim: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Audrey Root:** Writing – original draft, Data curation. **Michael C. Benson:** Writing – review & editing, Writing – original draft, Resources, Data curation, Conceptualization. **Daniel Beverly:** Writing – original draft, Resources, Data curation. **Jennifer.E. Johnson:** Writing – original draft, Visualization, Methodology, Conceptualization.

Manuel Lerdau: Writing – review & editing, Writing – original draft, Resources, Conceptualization. **Richard P. Phillips:** Resources, Investigation. **Kimberly A. Novick:** Writing – original draft, Resources, Investigation. **Xi Yang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Data curation, Conceptualization.

Declaration of competing interest

None declared.

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Appendix A. Quantum yields and symbols estimated using the PAM system

Table A1
Quantum yields and symbols estimated using the PAM system.

Symbols	Definition	Method
Φ_F	Quantum yield of Chlorophyll fluorescence	Eq. 1
Φ_P	Quantum yield of Photochemistry	Eq. 2
Φ_{NPQ}	Quantum yield of non-photochemical quenching	Eq. 3
$\Delta\Phi_F$	The response of Φ_F to changes in light intensity from low to high varies	Eq. 4
$\Delta\Phi_P$	The response of Φ_P to changes in light intensity from low to high varies	Eq. 4
$\Phi_{F_{angle}}$	Consideration of leaf angle effect on the quantum yield of chlorophyll fluorescence	Eq. 6
$\Phi_{F_{angle\&\alpha}}$	Consideration of leaf angle and absorbance effect on the quantum yield of chlorophyll fluorescence	Eq. 6
$\Phi_{P_{angle}}$	Consideration of leaf angle effect on the quantum yield of photochemistry	Eq. 6
$\Phi_{P_{angle\&\alpha}}$	Consideration of leaf angle and absorbance effect on the quantum yield of photochemistry	Eq. 6
Cyt b_6f conductance	Estimate of the rate of linear electron flow through the Cytochrome b_6f complex	Eq. 5
LEF	Linear electron flow	Eq. 5
Φ_{Pmax}	Maximum photochemical yield	Eq. S1
NPQ	Non-photochemical quenching	Eq. S1
q_L	The fraction of the 'open' Photosystem II (PS_{II}) reaction center	Eq. 5
F_S	Steady-state fluorescence	Observed
F'_M	Maximum fluorescence in the light	Observed
F'_O	Minimum fluorescence in the light	Observed
F_M	Maximum fluorescence in the dark	Observed
F_O	Minimum fluorescence in the dark	Observed

Appendix B. SCOPE simulations

We assumed that only direct light from the zenith direction was incident on the leaf and calculated intercepted light using a simplified $\cos(\textit{leaf angle})$ equation. In reality, diffuse light and backside illumination also contribute to leaf interception, potentially deviating from this assumption. To assess the impact of this simplification, we used SCOPE simulations to evaluate the effect of leaf angle on light interception and its relationship with Φ_P-Φ_F. Detailed simulation settings are provided in the Supplementary Information (Table S2).

The results showed that $\cos(\textit{leaf angle})$ captured more dynamic changes in intercepted light than SCOPE outputs (Fig. S21). $\frac{1+\cos(\textit{leaf angle})}{2}$ showed similar performance to SCOPE in estimating light interception. This equation $\left(\frac{1+\cos(\textit{leaf angle})}{2}\right)$ is widely used for simple estimations of diffuse light on the adaxial and abaxial surfaces of the leaf (Mousivand et al., 2015; Rakovec and Zakšek, 2012). Although it had a weaker influence on variation in light interception, it still contributed to the apparent Φ_P-Φ_F covariation across leaves (Fig. B1). When analyzed by species, the R^2 of oak increased from 0.43 (not significant) to 0.65 ($p < 0.1$), and that of maple increased from 0.34 (not significant) to 0.69 ($p < 0.05$) after considering leaf angle. This correlation became even stronger when absorption changes were considered (Fig. B1), indicating that a more vertical leaf angle combined with reduced absorption lowers intercepted light and significantly reinforces the Φ_P-Φ_F relationship. These results support our main conclusion that leaf angle variation makes the apparent Φ_P-Φ_F covariation across leaves more linear under realistic environmental conditions.

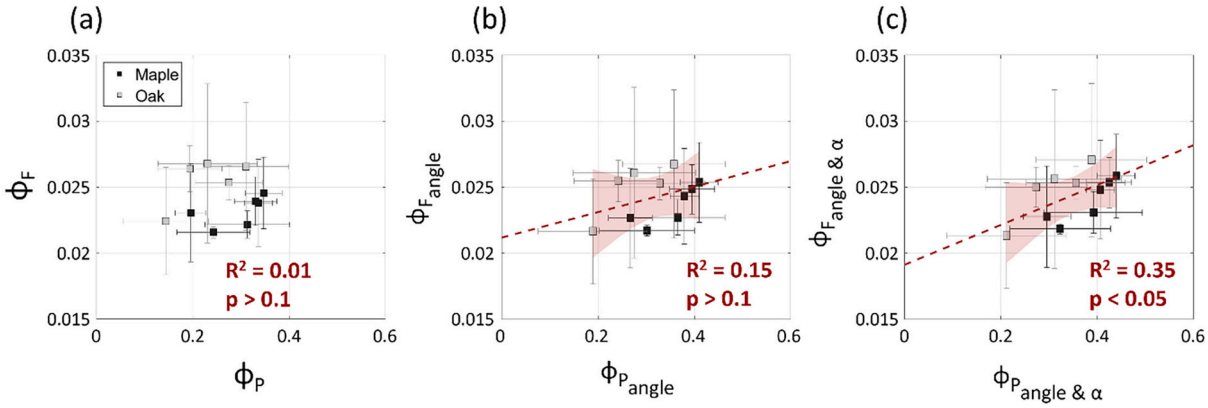


Fig. B1. (a) The relationship between Φ_P and Φ_F, at 650 μmol m^{-2} s^{-1}, (b) the correlation between angle-adjusted Φ_P and Φ_F; here, these values were estimated from measured light-response curves, using a value of PPFD that match what the leaves would intercept at a given leaf angle and light intensity, and (c) the correlation between angle-absorbance-adjusted Φ_P and Φ_F, again calculated as corresponding values of Φ_P or Φ_F from light response curves at the PPFD when considering the leaf angle and absorbance. To calculate the effect of leaf angle on intercepted light intensity, we used $\frac{1+cos(leaf\ angle)}{2}$, which produced results similar to the SCOPE simulation (See Fig. S21). Each data point with an error bar indicates the mean of light response data measured at each DOS with ±95% confidence intervals. The dashed red line indicates the linear regression line, and red shaded areas represent 95% confidence intervals. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Appendix C. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.rse.2026.115427.

Data availability

The data that support the findings of this study will be available in the Supporting Information and Github (https://kimhyodong.com/).

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